Calculating Kinetic Energy

Kinetic Energy=

WE can calculate the kinetic energy a moving object has if we know its mass and speed.

Kinetic energy= 1/2 of the objects Mass X the objects speed squared

$$E_k = \frac{1}{2}mv^2$$

Example

A rock has a mass of 2kg and is falling at a speed of 5 m/s, how much kinetic energy does it have?

$$E_k = \frac{1}{2}mv^2$$

$= (\frac{1}{2} \times 2) \times 5^2$

$= 1 \times 25$

$= 25$ J

Try this one

A toy car has a mass of 4kg and is rolling at a speed of 2 m/s, how much kinetic energy does it have?

$$E_k = \frac{1}{2}mv^2$$

= (½ x) X

= j

Comparing 3 Cars

Vehicle	Ford focus	Ford focus	Ford 4WD
Mass(kg)	1300	1300	2000
Speed(m/s)	10	20	10

Calculation	$E_k = \frac{1}{2}mv^2$ =(1/2 x) x ^{2}	$E_k = \frac{1}{2}mv^2$	$E_k = \frac{1}{2}mv^2$
Kinetic energy (j)			
Convert the Kinetic energy from Joules(j) to kilojoules(kJ) by dividing by 1000	/1000	/1000	/1000
Final Kj			

Draw a bar graph of these results.

Questions

Which vehicle had the most Kinetic energy?

Why was the ford focus able to have a greater kinetic energy than the larger 4wd, even though the 4wd had more Mass?

Why will it take twice as long to stop a car travelling at 65kph than a car travelling at 80kph?